

Macroeconomic Theory and Policy: Lecture W10 (Ch.15 and 18)

Quantitative DSGE Models

- Full DSGE models incorporate the complete dynamic response of all economic variables to all shocks
- The Smets–Wouters (2007) model is used by macroeconomic forecasters and central banks around the world as a building block
 - ▶ Incorporates shocks previously discussed
 - ▶ Includes both sticky prices and sticky wages and more
- One of the key advantages of this type of model is that it can be estimated
 - ▶ By “estimated,” we mean that the parameters of the DSGE model are chosen to make the model match the data for the U.S. economy as accurately as possible

Shocks

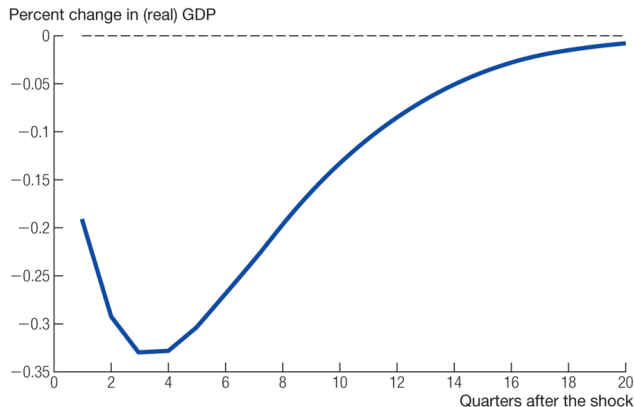
- Quantitative DSGE models incorporate multiple shocks simultaneously
- The Smets-Wouters model features:
 - ▶ Productivity shocks
 - ▶ Shocks to government purchases
 - ▶ Financial frictions
 - ▶ Investment shocks
 - ▶ Labour market shocks
 - ▶ Shocks to markups in product markets

Impulse Response Functions

- Impulse Response Functions (IRF) is one of the main tool to analyze the DSGE model (and multivariate time series model)
- The IRF shows how one macroeconomic variable of interest responds over time to an economic shock
- Each graph shows how each variables responses to a temporary shock
- The next slide shows the IRF for GDP in the estimated Smets-Wouters model
 - ▶ By what percent does GDP change after a temporary 1 percentage point increase in the fed funds rate?

The Response of GDP to a Monetary Policy Shock

- A 1 % increase in the fed funds rate causes GDP to fall by about 0.2 % in the first quarter after the shock
- At its peak three to four quarters after the shock, GDP is lower by 0.33 %

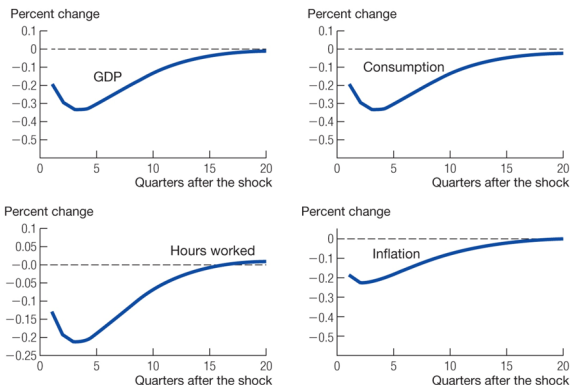


Source: Frank Smets and Rafael Wouters, "Shocks and Frictions in U.S. Business Cycles: A Bayesian DSGE Approach," *American Economic Review*, vol. 97, no. 3 (2007), pp. 586–606.

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The Dynamic Effects of a Monetary Policy Shock

- The DSGE models include many other variables, like consumption, hours worked, and inflation, allow us to study how each of these variables responds to a shock
- The response of the economy looks very much like the response we saw in our AS/AD framework

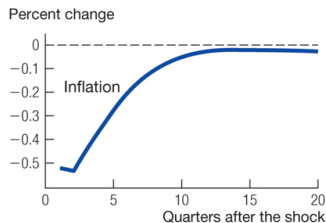
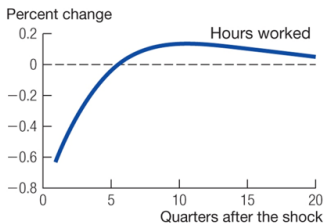
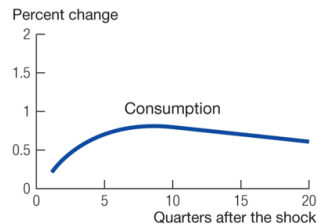
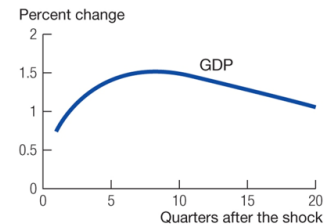


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The Dynamic Effects of a Monetary Policy Shock

- When there is a surprise tightening of monetary policy (a rise in interest rate), there are less investment
- As a result, less output to be produced which lower the consumption
 - ▶ Given the permanent income hypothesis, the consumption should varies less than the output
 - ▶ But Smets-Wouters's substantial adjustment costs to investment causes consumption to move by more (if the output is not being used in investment, it would go toward consumption)
 - ▶ Yet, a good model should not imply that the consumption move more than output
- As the output fall, according to the PC, the inflation rate should fall

The Dynamic Effects of a TFP Shock



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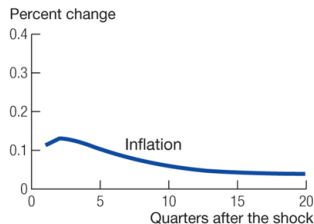
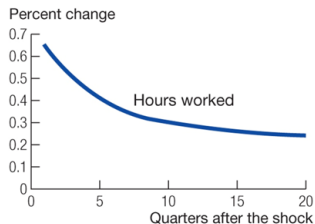
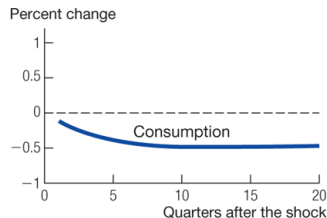
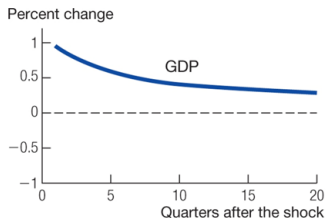
The Dynamic Effects of a TFP Shock

- The TFP shocks considered here decay over time instead of disappear entirely in one period
- As the positive TFP shocks arrive, the GDP rises
 - ▶ The increase is even higher than the the increase in the TFP (i.e., 1%)
 - ▶ This is because of the capital accumulation, which is similar to the case discussed in the Solow model
- Given the permanent income hypothesis, individual prefer to smooth consumption.
 - ▶ The consumption increases less than the increase in output

The Dynamic Effects of a TFP Shock

- The hour worked may look puzzling, as the stylized model would suggest an increase in both wage and employment
 - ▶ This model feature sticky prices, the firm supply as many as amount demanded
 - ▶ Since the consumption rises by little, the rise in TFP is more than enough to supply the amount demanded in the economy
- The TFP shocks also raises potential GDP and expands supply in the economy
 - ▶ In the AS/AD model, this shifts down the AS curve
 - ▶ However, the actual output doesn't increase as much as the shock initially, the actual output would be lower than the potential output temporary
 - ▶ Resulting a fall in inflation rate

The Dynamic Effects of a Shock to Government Purchases

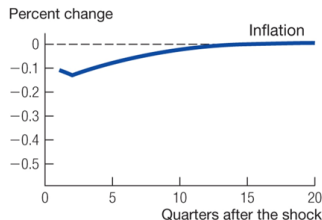
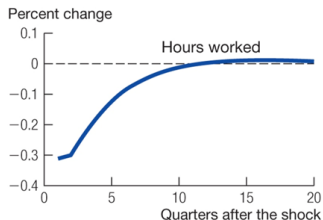
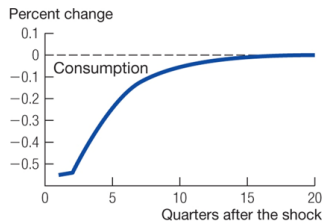
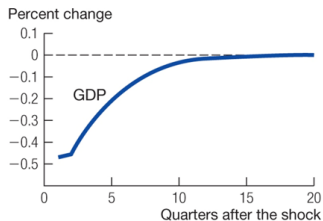


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The Dynamic Effects of a Shock to Government Purchases

- The stylized model suggests that the wages would fall while the hours worked increases
 - ▶ As the government increases its spending, consumer would expect an increase in tax at some point
 - ▶ The wealth decrease and lead to a decrease in consumption
 - ▶ The labour supply shift to the right
- The model generate similar response as expected
- Since the labour supply increases, the effect on inflation is little

The Dynamic Effects of a Financial Frictions Shock



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The Dynamic Effects of a Financial Frictions Shock

- This particular financial shock implemented as a wedge between the rate of saving and borrowing, because of asymmetric information problem
 - ▶ Implies it has the largest effect on consumption
- Otherwise, the shock looks very similar to a monetary policy shock, as both work through interest rates

Conclusion

- Macroeconomists argue about the importance of various shocks and various frictions in accounting for economic fluctuations
 - ▶ These arguments often center around the extent to which certain frictions are quantitatively important
 - ★ How sticky are wages and prices?
 - ★ How important are credit market frictions?
 - ★ How elastic is labor supply in the economy?

Conclusion

- The basic framework we should use to understand macroeconomic fluctuations developed over the past 50 years
 - ▶ A key element is a basic growth model like the Solow model
 - ▶ Including more microfoundation elements
- The basic DSGE framework has been enriched in many ways recently
- The models make precise quantitative predictions about the complete dynamics of a host of endogenous variables

What are You Going to Learn (Ch. 18)

- Government spending, taxation, budget deficits, and the debt-GDP ratio nationally and globally
- The government's intertemporal budget constraint
- The economic consequences of budget deficits
- The fiscal problem of the twenty-first century: how to finance rising health expenditures







































Next Lecture

- Next lecture will Ch.10